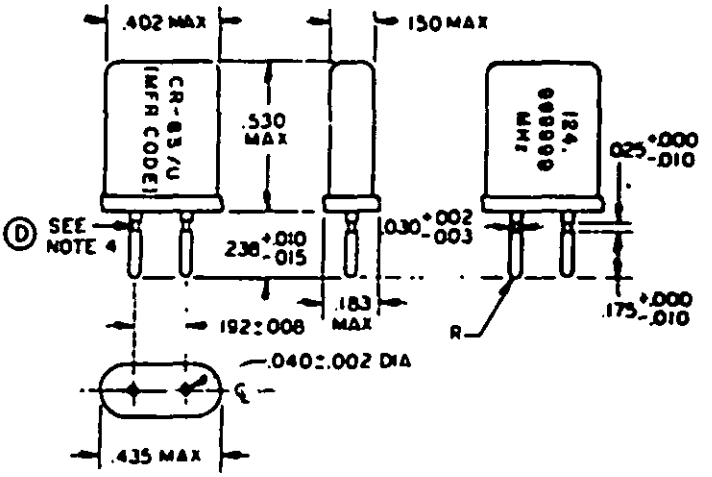
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-83/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 50 to 125 MHz; fifth overtone; noncontrolled; series.



INCHES	MM	INCHES	MM
.002	.05	.150	3.81
.003	.08	.175	4.45
.008	.20	.183	4.65
.010	.25	.192	4.88
.015	.38	.238	6.05
.025	.64	.402	10.21
.030	.76	.435	11.05
.040	1.02	.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designation, frequency, and code designating characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.
(D) 4. The pin undercut may be omitted.

(D) denotes change

FSC 5955

Source: https://assist.dla.mil -- Downloaded: 202610-16T16:36Z
Check the source to verify that this is the current version before use.

TABLE 1. Requirements.

Requirement	Specified value or condition
Holder	HC-25/U (MIL-H-10056/11)
Dimensions, marking, and configuration	See figure
Frequency range	50 to 125 MHz, inclusive
Capacitance, shunt	7 pF, maximum
Frequency tolerances:	
Primary operating temperature range	±20 PPM
Secondary operating temperature range	±30 PPM
Equivalent resistance	MHz — Ohms, maximum 50 to 90 — 50 90+ to 125 — 60
Mode of oscillation	Fifth mechanical overtone
Government reference standard test sets	TS-683/TSM or AN/TSM-15
Calibration values:	
Resistance	60 ohms
Resistor voltage drop	0.35 volt
Drive level, rated	2.0 ± 0.4 mW
Operating temperature ranges:	
Primary	-40° to +90°C, inclusive
Secondary	-55° to -40° and +90° to +105°C, inclusive
Shock (specified pulse):	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Vibration high frequency: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Temperature run:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	5 PPM

1/ Method 204 of MIL-STD-202, test condition A.

NOTE: NATO guidance type NXT54.

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - EL, MI, MU, SG, SM
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5955-0408-33)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:36Z
Check the source to verify that this is the current version before use.